

Patent Management System - API Documentation

Applicant Interface

The Applicant Interface in the frontend consists of five main tabs:

1. **Dashboard:** Contains an introduction to PMS, workflow overview, standard documents, and navigation cards.
 - **API Requirement:** No API needed.
2. **Submit Patent (IP Filing Form):** Allows applicants to fill and submit the patent filing form. They can save drafts before submission.
 - **API Requirement:** Required for submitting forms and saving drafts.
3. **Submitted Applications:** Displays submitted applications along with their current status.
 - **API Requirement:** Required to fetch submitted applications and their status.
4. **Saved Drafts:** Allows applicants to view saved drafts and continue filling them until submission.
 - **API Requirement:** Required to fetch and update saved drafts.
5. **Notifications:** Displays notifications about status updates, or new messages from PCC Admin or Director.
 - **API Requirement:** Required to fetch notifications in real-time.

1. Applicant API Endpoints

1.1 Submit Patent Application

- **Endpoint:** `patentsystem/applicant/applications/submit`
- **Description:** Allows an applicant to submit a new patent application.
- **Input Parameters:**
 - `title` (string, required): Title of the patent
 - `inventors` (array, required): List of inventors with the following structure:

```
[
  {
    "name": "Inventor-1",
    "email": "inventor.1@example.com",
    "contact_address": "1234 abc",
    "mobile_number": "+1234567890"
  },
  {
    "name": "Inventor-2",
    "email": "inventor.2@example.com",
    "contact_address": "5678 Oak",
    "mobile_number": "+1234567890"
  }
]
```

- **area_of_invention** (string, required): Field of the invention
- **problem_statement** (string, required): Problem in the area of invention
- **objective** (string, required): Objective of the invention
- **novelty** (string, required): Novelty of the invention
- **advantages** (string, required): Advantages over existing patents/literature
- **tested_experimentally** (boolean, required): Whether the invention has been tested
- **proof_of_concept** (file, optional): Supporting document for proof of concept
- **applications** (string, required): Possible applications of the invention
- **significant_use_of_funds** (string, required): Whether funds from the institution were used
- **funding_source** (string, required): Source of funding (if applicable)
- **funding_agency_details** (file, optional): Upload of funding agreement (if applicable)
- **publication_details** (string, optional): Details of prior presentations/publications
- **sponsored_research** (string, optional): Research agreement details (if applicable)
- **mou_details** (file, optional): Upload of MOU (if applicable)
- **academic_research** (string, optional): Academic research involvement
- **inventor_contributions** (array, required): List of inventors with percentage contribution for revenue sharing
- **target_companies** (array, optional): List of target companies with contact details
- **development_stage** (string, required): Current stage of development (Embryonic, Partially developed, Off-the-shelf)
- **signed_form** (file, required): Upload of duly signed Form-III
- **user_id** (integer, required): ID of the applicant

- **Expected Output:**

```
{
  "message": "Application submitted successfully",
  "application_id": 12345,
  "token": "IIITDMJ/AGR/20241107/001/AAS/104"
}
```

- **Status:** Implemented

1.2 View Application(s)

- **Fetch Applications by User ID**

- **Endpoint:** `/patentsystem/applicant/applications/`
- **Method:** `GET`

- **Description:** Retrieves all applications associated with a given applicant based on their user ID.

- **Input Parameters:**

- **user_id** (integer, required): Unique ID of the user

- **Expected Output:**

```
{
  "applications": [
    {
      "application_id": 101,
      "title": "AI-Based Patent Filing",
      "token_no": "AIPT2025001",
      "attorney_name": "John Doe",
      "submitted_date": "2025-03-10"
    },
    {
      "application_id": 102,
      "title": "Blockchain Security Patent",
      "token_no": "BSPT2025002",
      "attorney_name": "Not Assigned",
      "submitted_date": "2025-02-28"
    }
  ]
}
```

- **Status:** Implemented

- **Fetch Application Details by application ID**

- **Endpoint:**

`patentsystem/applicant/applications/application_details`

- **Method:** GET

- **Description:** Fetches details of a specific application along with its status.

- **Input Parameters:**

- **application_id** (integer, required): Unique ID of the application

- **Expected Output:**

```
{
  "application_id": 12345,
  "title": "AI-Based Patent Filing",
```

```
"status": "Under Review",
"last_updated": "2025-02-22"
}
```

- **Status:** Pending implementation

1.2 Track Application Status

- **Endpoint:** `/api/applicant/status/{application_id}`
- **Description:** Fetches the current status of a patent application.
- **Input Parameters:**
 - `application_id` (integer, required): Unique ID of the application
- **Expected Output:**

```
{
  "application_id": 12345,
  "status": "Under Review",
  "assigned_attorney": "John Doe"
}
```
- **Status:** Pending implementation

1.3 View Saved Drafts

- **Endpoint:** `/api/applicant/drafts`
- **Description:** Fetches the list of saved drafts.
- **Input Parameters:**
 - `user_id` (integer, required): ID of the applicant
- **Expected Output:**

```
{
  "drafts": [
    {
      "draft_id": 6789,
      "title": "AI-Based Patent Filing",
      "last_updated": "2025-02-22"
    }
  ]
}
```
- **Status:** Pending Implementation

1.4 Notifications

- **Endpoint:** `/api/applicant/notifications`
- **Description:** Fetches notifications related to application status updates or messages from PCC Admin/Director.
- **Input Parameters:**
 - `user_id` (integer, required): ID of the applicant

- **Expected Output:**

```
{
  "notifications": [
    {
      "message": "Your application status has been updated to 'Reviewed'",
      "timestamp": "2025-02-22T10:00:00Z"
    }
  ]
}
```

- **Status:** Pending implementation

1.5 Fetch a Specific Draft

- **Endpoint:** `/api/applicant/draft/{draft_id}`
- **Description:** Fetches details of a specific saved draft.
- **Input Parameters:**

```
{
  "draft_id": 6789
}
```

- **Expected Output:**

```
{
  "draft_id": 6789,
  "title": "AI-Based Patent Filing",
  "inventors": [
    {
      "name": "John Doe",
      "email": "john.doe@example.com",
      "contact_address": "123 Main St, City, Country",
      "mobile_number": "+1234567890"
    }
  ],
  "area_of_invention": "Artificial Intelligence",
  "problem_statement": "Lack of an automated patent filing system.",
  "last_updated": "2025-02-22T12:00:00Z"
}
```

PCC Admin Interface

The PCC Admin Interface in the frontend consists of five main tabs:

1. Dashboard

- Contains an overview of patent applications submitted each year, including the number of applications in different statuses (submitted, reviewed, patented, etc.).

- **API Requirement:** Required to fetch yearly application statistics for display.
2. New Applications
 - Displays newly submitted applications in a table format with columns: Token Number, Patent Title, Submitted By, Designation, Department, Date-Time, and View. Clicking on "View" opens the full application details.
 - A remarks textbox is provided for PCC Admin to add comments. A status bar indicates the application's progress in the workflow.
 - **API Requirement:** Required to fetch new applications, update remarks, and manage application statuses.
 3. View Applications
 - Allows PCC Admin to view the details and current status of any application at any stage of the workflow. PCC Admin can also update the application's status.
 - **API Requirement:** Required to fetch application details and update the status.
 4. Manage Attorney Assignment
 - Lists attorneys with their Name, Attorney ID, and actions to view assigned applications, assign applications to attorneys, and add or remove attorneys.
 - **API Requirement:** Required to fetch, assign, and manage attorneys.
 5. Notifications
 - Displays notifications related to new applications, application status updates, and communication with applicants or directors.
 - **API Requirement:** Required to fetch and send notifications in real-time.

2. PCCAdmin API Endpoints

2.3 View New Applications

- **Endpoint:** `/api/pccAdmin/new-applications`
- **Method:** GET
- **Description:** Fetches a list of newly submitted applications awaiting review.
- **Input Parameters:** None
- **Expected Output:**
 - {
 - "applications": [
 - {
 - "token": "IIITDMJ/AGR/20241107/001/AAS/104",
 - "title": "AI-Based Patent Filing",
 - "submitted_by": "Jane Doe",
 - "designation": "Professor",
 - "department": "Computer Science",
 - "date_time": "2025-02-22T10:00:00Z"
 - }
 -]
 - }
- **Status:** Not started

2.4 View Specific Application Details

- **Endpoint:** `/api/pccAdmin/application/{application_id}`
- **Method:** GET
- **Description:** Fetches the details of a specific patent application.
- **Input Parameters:**
 - `application_id` (integer, required): Unique ID of the application
- **Expected Output:**
 - {
 - "application_id": 12345,
 - "title": "AI-Based Patent Filing",
 - "submitted_by": "Jane Doe",
 - "status": "Under Review",
 - "details": {
 - "area_of_invention": "Artificial Intelligence",
 - "problem_statement": "Lack of an automated patent filing system."
 - }
 - }
- **Status:** Not started

2.5 Add Remarks to Application

- **Endpoint:** `/api/pccAdmin/application/{application_id}/remarks`
- **Method:** POST
- **Description:** Allows PCC Admin to add remarks to a patent application.
- **Input Parameters:**
 - `application_id` (integer, required): Unique ID of the application
 - `remarks` (string, required): Remarks or comments from the PCC Admin
- **Expected Output:**
 - {
 - "message": "Remarks added successfully"
 - }
- **Status:** Pending implementation

2.6 Update Application Status

- **Endpoint:** `/api/pccAdmin/application/{application_id}/status`
- **Method:** PATCH
- **Description:** Updates the status of an application in the workflow.
- **Input Parameters:**
 - `application_id` (integer, required): Unique ID of the application
 - `status` (string, required): New status of the application
- **Expected Output:**
 - {
 - "application_id": 12345,
 - "status": "Attorney Assignment",
 - "message": "Application status updated successfully"

- }
- **Status:** In progress

2.7 Manage Attorney Assignments

- **Endpoint:** `/api/pccAdmin/manage-attorneys`
- **Method:** GET
- **Description:** Fetches the list of attorneys and their assigned applications.
- **Input Parameters:** None
- **Expected Output:**
 - {
 - "attorneys": [
 - {
 - "attorney_id": 1,
 - "name": "John Doe",
 - "assigned_applications": 5
 - }
 -]
 - }
- **Status:** Not started

2.8 Assign Application to Attorney

- **Endpoint:** `/api/pccAdmin/assign-application`
- **Method:** POST
- **Description:** Assigns a patent application to an attorney.
- **Input Parameters:**
 - `application_id` (integer, required): Unique ID of the application
 - `attorney_id` (integer, required): ID of the attorney
- **Expected Output:**
 - {
 - "message": "Application assigned to attorney successfully"
 - }
- **Status:** Pending implementation

2.9 Remove Attorney Assignment

- **Endpoint:** `/api/pccAdmin/remove-attorney`
- **Method:** DELETE
- **Description:** Removes an attorney from an assigned application.
- **Input Parameters:**
 - `application_id` (integer, required): Unique ID of the application
 - `attorney_id` (integer, required): ID of the attorney
- **Expected Output:**
 - {
 - "message": "Attorney removed successfully"
 - }

- **Status:** Not started

2.10 Notifications

- **Endpoint:** `/api/pccAdmin/notifications`
- **Method:** GET
- **Description:** Fetches notifications related to application updates.
- **Input Parameters:** None
- **Expected Output:**
 - {
 - "notifications": [
 - {
 - "message": "New application submitted: AI-Based Patent Filing",
 - "timestamp": "2025-02-22T10:00:00Z"
 - }
 -]
 - }
- **Status:** Pending implementation

2.1 Assign Attorney to Application

- **Endpoint:** `/api/pccAdmin/assign-attorney`
- **Method:** POST
- **Description:** Assigns an attorney to review a patent application.
- **Input Parameters:**
 - `application_id` (integer, required): Unique application ID
 - `attorney_id` (integer, required): ID of the attorney
- **Expected Output:**
 - {
 - "message": "Attorney assigned successfully",
 - "attorney": "John Doe"
 - }
- **Status:** Not started

2.2 Generate Search Report

- **Endpoint:** `/api/pccadmin/generate-search-report/{application_id}`
- **Method:** POST
- **Description:** Generates a patent search report for the given application.
- **Input Parameters:**
 - `application_id` (integer, required): Unique ID of the application
- **Expected Output:**
 - {
 - "application_id": 12345,
 - "report_status": "Generated",
 - "download_link": "<https://pms.com/reports/12345.pdf>"
 - }

- }
 - **Status:** Pending implementation
-

Director Interface

The **Director Interface** in the frontend consists of four main tabs:

1. Dashboard

- Similar to the PCC Admin dashboard, it provides an overview of patent applications categorized by status (submitted, under review, patented, etc.), along with yearly statistics.
- **API Requirement:** Required to fetch yearly application statistics for display.

2. View New Applications

- Displays applications forwarded by PCC Admin for review, allowing the director to assess and take necessary actions.
- **API Requirement:** Required to fetch forwarded applications.

3. Status of Applications

- Allows the director to view the details and current status of all applications at different workflow stages.
- **API Requirement:** Required to fetch application details and statuses.

4. Notifications

- Displays notifications related to new application arrivals, application status updates, and messages from PCC Admin or system-generated alerts.
- **API Requirement:** Required to fetch and send notifications in real-time.

3. Director API Endpoints

3.1. Fetch Dashboard Overview

- **Endpoint:** `/api/director/dashboard-overview`
- **Method:** `GET`
- **Description:** Fetches an overview of applications by year, showing statistics like total submitted, reviewed, patented, and pending applications.
- **Input Parameters:** None
- **Expected Output:**

```
{
  "yearly_overview": [
    {
      "year": 2024,
      "total_applications": 120,
      "reviewed": 85,
      "patented": 30,
      "pending": 25
    },
    {
      "year": 2023,
      "total_applications": 105,
```

```

        "reviewed": 90,
        "patented": 40,
        "pending": 15
    }
]
}

```

- **Status:** Pending Implementation

3.2. Fetch New Applications Forwarded by PCC Admin

- **Endpoint:** `/api/director/new-applications`
- **Method:** GET
- **Description:** Retrieves the list of new applications forwarded by PCC Admin for review.
- **Input Parameters:** None
- **Expected Output:**

```

{
  "applications": [
    {
      "application_id": 12345,
      "title": "AI-Based Patent Filing",
      "submitted_by": "Alice Johnson",
      "department": "Computer Science",
      "submission_date": "2025-02-20",
      "status": "Forwarded by PCC Admin"
    },
    {
      "application_id": 12346,
      "title": "Green Energy Storage",
      "submitted_by": "Bob Smith",
      "department": "Electrical Engineering",
      "submission_date": "2025-02-22",
      "status": "Forwarded by PCC Admin"
    }
  ]
}

```

- **Status:** Pending implementation

3.3. Fetch Application Details and Status

- **Endpoint:** `/api/director/application/{application_id}`
- **Method:** GET
- **Description:** Retrieves details and the current status of a specific application.
- **Input Parameters:**
 - `application_id` (integer, required): Unique application ID
- **Expected Output:**

- ```
{
 "application_id": 12345,
 "title": "AI-Based Patent Filing",
 "submitted_by": "Alice Johnson",
 "department": "Computer Science",
 "submission_date": "2025-02-20",
 "status": "Under Review",
 "assigned_attorney": "John Doe",
 "last_updated": "2025-02-22"
}
```
- **Status:** Not started

### 3.4. Fetch Notifications

- **Endpoint:** `/api/director/notifications`
  - **Method:** `GET`
  - **Description:** Fetches notifications regarding new applications, status updates, or messages from PCC Admin.
  - **Input Parameters:**
    - `user_id` (integer, required): ID of the director
  - **Expected Output:**

```
{
 "notifications": [
 {
 "message": "New application 'AI-Based Patent Filing' has been forwarded for review.",
 "timestamp": "2025-02-22T10:00:00Z"
 },
 {
 "message": "Application 'Green Energy Storage' has been updated to 'Patentability Check'.",
 "timestamp": "2025-02-22T12:30:00Z"
 }
]
}
```
  - **Status:** Pending implementation
-